A Computational Virtual Patient Pipeline for Predicting Perceptual Capabilities with Visual Prostheses

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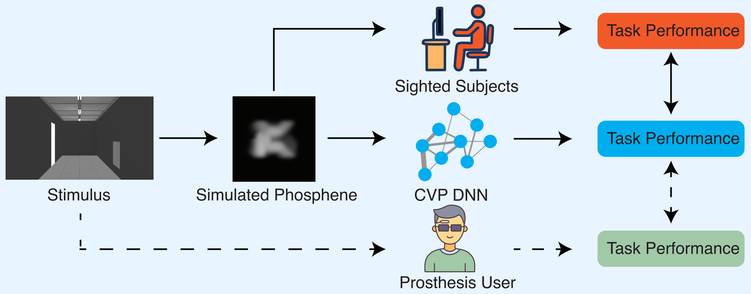
VIU vision o image understanding

Why Simulate Prosthetic Perception?

- Current visual prostheses elicit rudimentary phosphenes that differ significantly from natural vision
- Users are typically evaluated post-implantation using the Functional Low-Vision Observer Rated Assessment (FLORA) (Geruschat et al., 2015)
- A fundamental challenge is predicting perceptual capabilities before implantation

Computational Virtual Patient (CVP) Framework



Simulating and Evaluating Perception with Prosthetic Vision

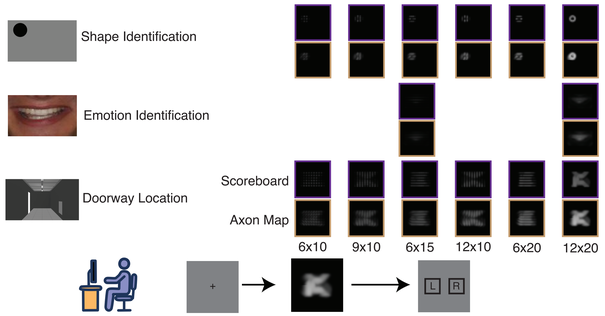
Tasks: Shape identification, emotion identification, person facing direction, doorway location, person location, window location

Phosphene Simulation:

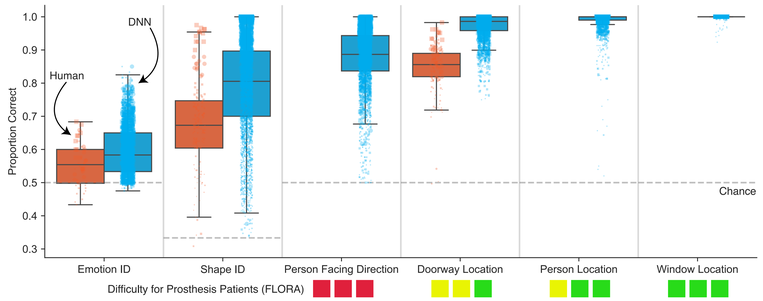
- Two phosphene generation models: the 'scoreboard' model (producing localized points of light) and the 'axon map' model (which incorporates the anatomical layout of retinal ganglion cell axons) (Beyeler et al., 2017, 2019)
- Six electrode configurations: 6×10, 9×10, 6×15, 12×10, 6×20, and 12×20

Evaluation:

- Human observers in a forced-fixation, forced-choice task
- Deep neural networks (DNNs) trained to perform the same tasks: ResNet, VGG, EfficientNet, RegNetY, MobileNetV2, Vision Transformers (ViT), and ConvNeXt
- We compared human and DNN performance across all tasks, electrode configurations, and phosphene models

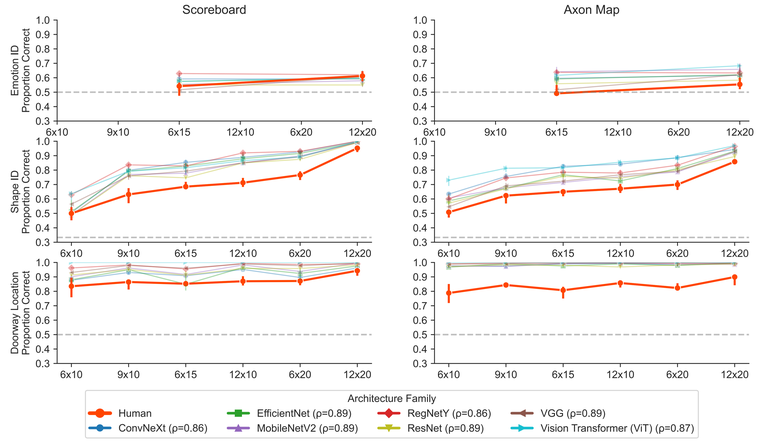


Human–Model Agreement Across Visual Tasks



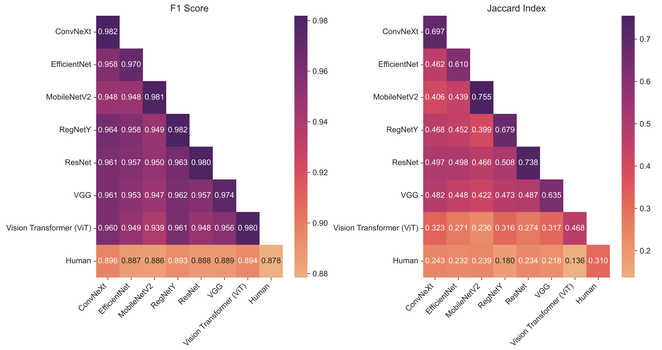
- DNNs tended to outperform human participants
- Human observers and DNNs displayed similar performance variation across tasks
- Task-level performance rankings were consistent between human psychophysics and DNNs
- FLORA task difficulty assessments were preserved in DNN performance patterns

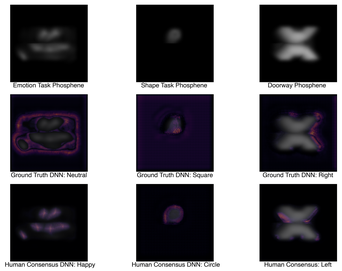
Shared Sensitivity to Implant Configurations



- Humans and DNNs showed strong agreement in performance across implant designs (Spearman's ρ = 0.86 to 0.89)
- Human–DNN alignment persisted across both 'scoreboard' and 'axon map' models, and across DNN architectures, despite perceptual distortions
- Performance generally increased with higher electrode counts
- In both humans and DNNs, shape task performance declined under the 'axon map' model for high-density configurations (e.g., 12×20, 6×20, 12×10; Δ = -0.042 to -0.094; $p < 0.05$)

Trial-Level Agreement and Visual Attribution





- **F1 score** (agreement on correct classifications) was consistently high between humans and DNNs:

$$F1 = \frac{2 \cdot (\text{both correct})}{2 \cdot (\text{both correct}) + (\text{A wrong, B correct}) + (\text{A correct, B wrong})}$$

- **Jaccard index** (overlap in misclassified items) revealed distinct misclassification patterns:

$$\text{Jaccard} = \frac{(\text{both wrong})}{(\text{A wrong}) \cup (\text{B wrong})}$$

- **Attribution analysis**: Separate CNNs trained to predict (i) ground truth and (ii) human choices revealed divergent saliency patterns in cases of disagreement, suggesting reliance on different perceptual cues

Key Takeaways

Computational Virtual Patients (CVPs) offer a powerful framework for predicting task difficulty and assessing the impact of electrode configuration on performance. Our results highlight their potential as a tool for guiding task-specific evaluation and enhancing visual prosthesis design.

Acknowledgements

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References

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